

Project Stardust

AlforE Subscriber Plugin: data flow, marketplace workflow and actionable insights

Overview

Stardust is a universal, open-source plugin that meters AI usage (tokens, cost per service and environmental impact) wherever AI is consumed: inside AI chat apps, in the major browsers, and at the infrastructure level. The same measurement feeds three outcomes: subscriber ROI dashboards, the AlforE carbon sequestration marketplace, and public sustainability benchmarks.

1. Data flow: from capture to insight

Usage is captured at the point of creation, metered into a common record, enriched with cost and impact, aggregated, and delivered to the people who can act on it.

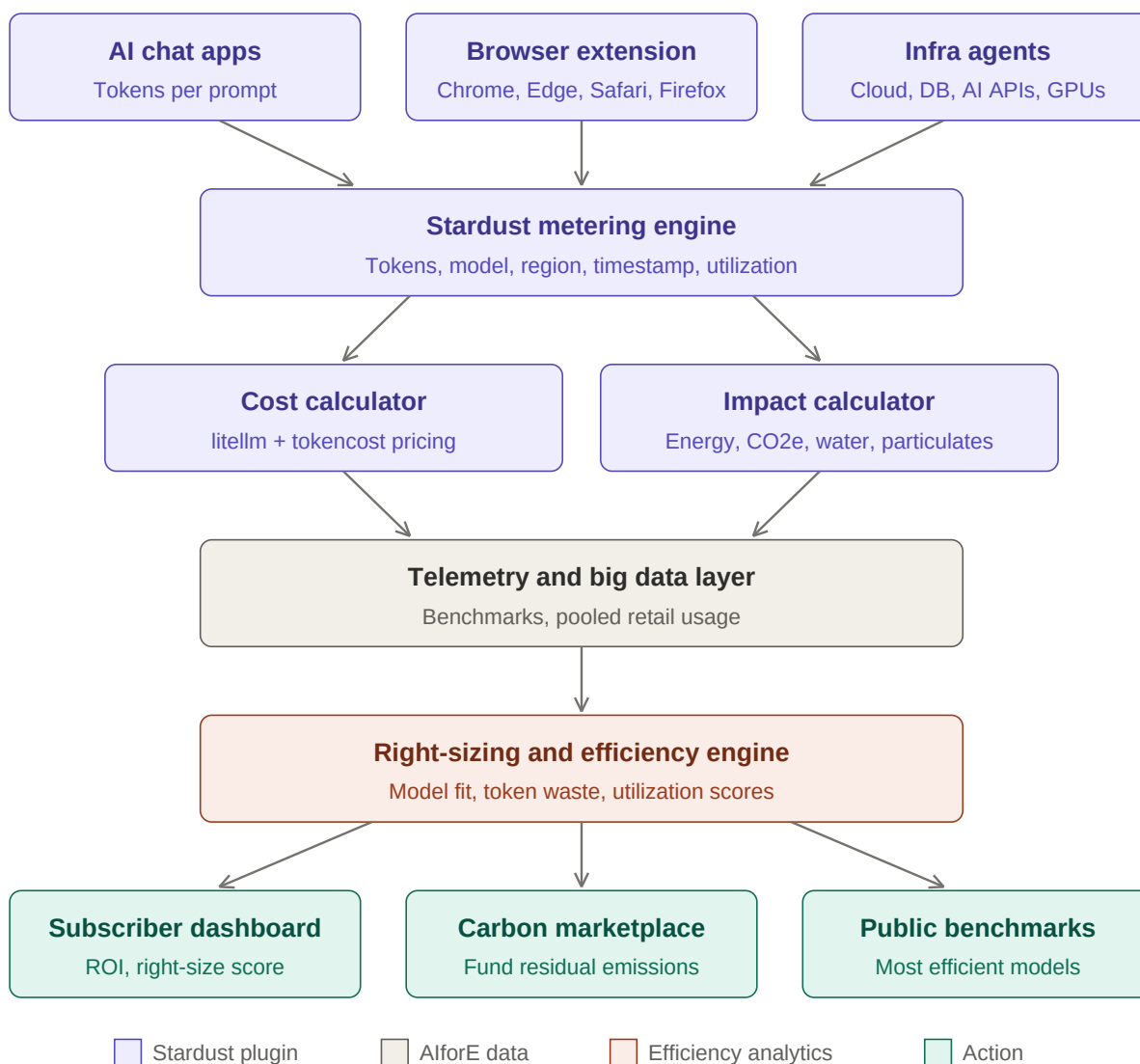


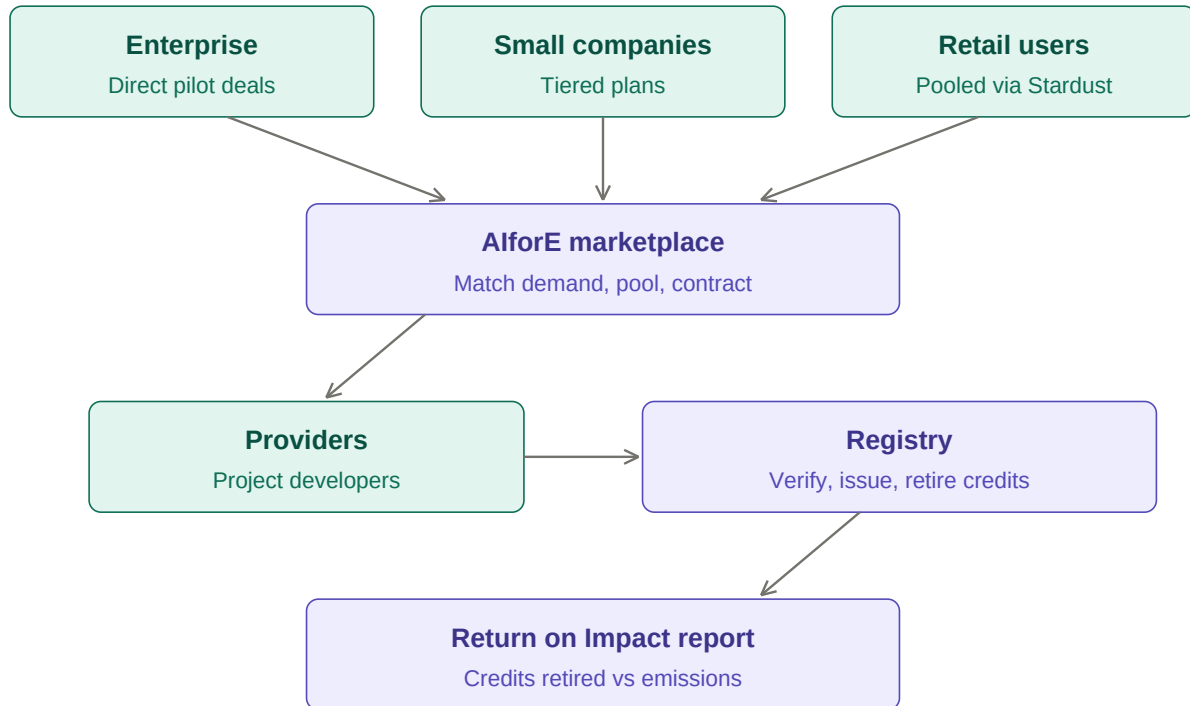
Figure 1. Stardust end-to-end data flow

Stage by stage

- **Capture.** Plugins in AI chat apps (Claude, ChatGPT, Gemini and others), browser extensions (Chrome, Edge, Safari, Firefox) and infrastructure agents (cloud, databases, AI APIs) record each AI call where it happens.
- **Meter.** The Stardust engine normalizes every event into one record: tokens in and out, model, provider, region and timestamp.
- **Enrich.** The cost calculator applies pricing from BerriAI/litellm and AgentOps-AI/tokencost (MIT-licensed). The impact calculator converts usage into electricity, energy, greenhouse gases (CO2e), water and particulate waste.
- **Aggregate.** The Sustainable AI Telemetry and Big Data layer stores records, builds benchmarks and pools retail usage for the marketplace.
- **Analyze.** The right-sizing and efficiency engine scores each workload for model fit, token waste and infrastructure utilization (detailed in Section 3).
- **Act.** Results surface as subscriber dashboards, marketplace purchases for residual emissions, and public benchmark reports.

2. Carbon marketplace workflow

Subscribers (buyers) enter at three tiers. AlforE matches and pools demand, Providers (sequestration project developers) deliver, and the registry verifies, issues and retires credits. The resulting Return on Impact report closes the loop.



Loop: report feeds back to subscriber dashboards and next-quarter purchasing

Figure 2. Subscriber / Provider marketplace workflow

3. Measuring right-sizing and efficiency

Because the telemetry stack sees every call with its model, tokens, cost, energy and timing, it can answer two questions: is each workload running on the right-sized model and infrastructure, and how much useful output does each dollar and kilowatt-hour produce?

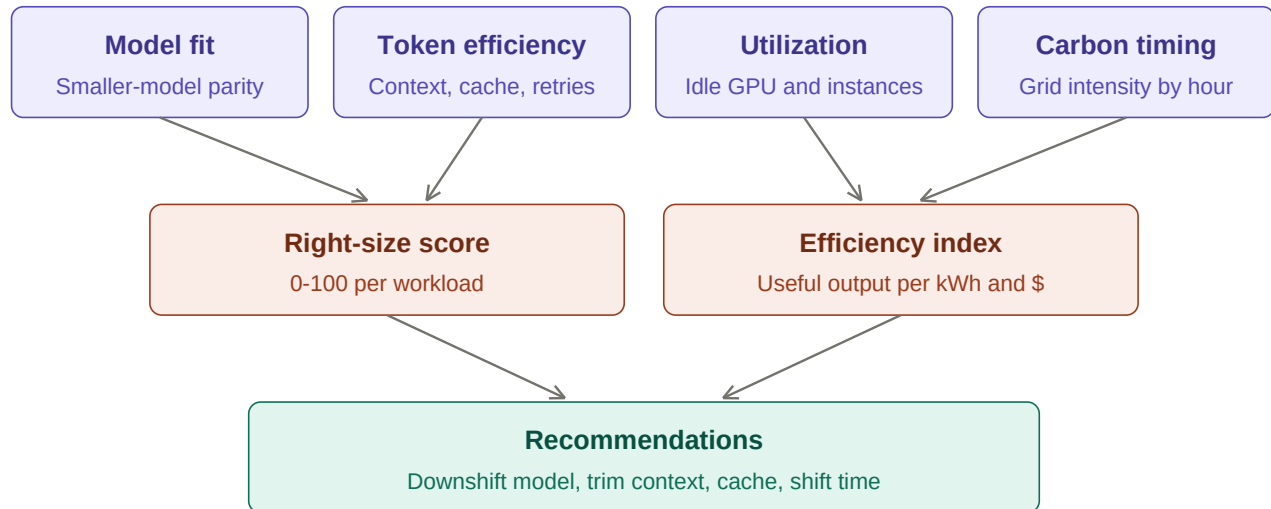


Figure 3. How the telemetry stack measures right-sizing and efficiency

Signals

- **Model fit.** A sample of prompts is shadow-tested on a smaller or cheaper model. If output quality holds (parity score above a set threshold), the workload is oversized.
- **Token efficiency.** Input-to-output token ratio, repeated system prompts and context bloat, prompt-cache hit rate, and retry or regenerate rate. High values signal paid-for tokens that produce nothing.
- **Utilization.** From infrastructure agents: GPU and instance utilization, idle provisioned capacity, and batchable work running in real time.
- **Carbon timing.** Grid carbon intensity at the time and region each job ran, compared with the cleanest available window.

Scores

- **Right-size score (0-100)** per workload, team or application: how closely the model and infrastructure match what the task actually needs.
- **Efficiency index:** useful output (completed tasks or accepted responses) per dollar, per kWh and per gram of CO₂e, trended over time.

Why it matters

Right-sizing cuts cost and emissions at the source; the carbon marketplace then offsets only what remains. Together they give Subscribers a reduce-first, offset-second story that holds up under ESG scrutiny.

4. Actionable insights

What the pipeline produces, organized by who acts on it.

Subscribers (enterprise and small companies)

- **Right-size score and savings estimate** per workload, showing exactly which calls can move to a smaller model, a trimmed prompt or a cleaner time window.
- **Cost per task and per model**, so teams can move work to a cheaper model where quality holds.
- **Carbon, energy and water per workload**, so jobs can shift to cleaner regions or off-peak hours.
- **Offset gap** (emissions measured vs. credits retired), which sets how much sequestration to buy next quarter.
- **Return on Impact**: the dollar value of credits alongside AI ROI, ready for ESG and CSRD-style disclosure.

Retail users

- **A live footprint indicator per chat**, plus a one-tap contribution to the pooled sequestration fund.

Providers

- **Demand forecasts** from aggregated usage trends, so they can size projects and price credits ahead of buyers.

AlforE and the public

- **Benchmarks ranking models and providers** by cost, impact and efficiency per useful output, useful as grant material, research output and pressure on vendors to disclose.

Key design point

One measurement (tokens, model, region, time) serves all three outcomes. Getting the capture schema right in the next version of the product spec means every downstream dashboard, credit purchase and benchmark inherits it.